

The Art of R

TM the Great Unread

DTL|Digital Arts Initiative
Interacting Minds Centre|Aarhus University

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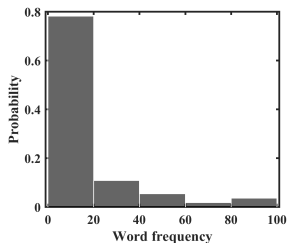


Words

Words are (one of) the basic units of meaning

Most TM techniques rely on word frequencies, that is, we tokenize a text at the word level and count the number of tokens for each type.

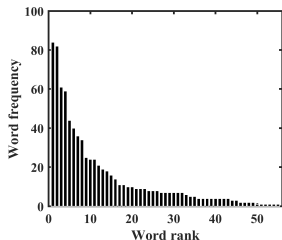
I am Daniel	'I' 'am' 'Daniel' 'I'	a	1	59	0.073
I am Sam	'am' 'Sam' 'Sam'	am	1	16	0.02
Sam I am	'I' 'am' 'That'	and	1	24	0.03
That Sam-I-am	'Sam' 'I' 'am'	anywhere	1	1	0.001
That Sam-I-am!	'That' 'Sam' 'I'	anywhere	1	7	0.009
I do not like	'am' 'I' 'do' 'not'	...			
that Sam-I-am	'like' 'that' 'Sam'	you	1	34	0.042
...	'I' 'am' ...	<i>total</i>	55	804	1.0



Most words are infrequent, but a few words are very frequent

'i' 'not' 'them' 'a' 'like' 'in' 'do'
'you' 'would'

Model a text as a distribution over words. Some words are more likely than other.



Often times we are looking at the mid-range (not too likely and not too unlikely).

Word frequencies

Binary term frequency: $f_{t,d} = 0, 1$

Raw term frequency: $f_{t,d} = N(t, D)$

Normalized term frequency ¹ $f_{t,d} = \frac{N(t, D)}{N(D)}$

IDF weighted term frequency ²: $tfidf(t, d, D) = f_{t,d} \cdot \log \frac{|D|}{|\{d \in D : t \in d\}|}$

¹prevents bias towards longer documents

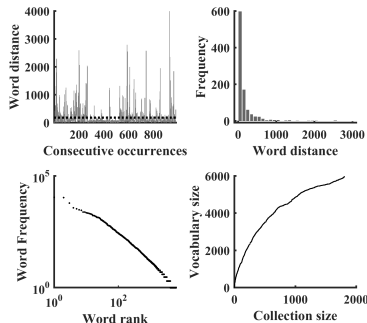
²removes non-informative words

Fundamental properties of language

Words occur in bursts - if word occurs it is likely to occur again in close proximity

A word's frequency is inversely proportional to its rank

The vocabulary increases as a function of the number of texts, but the increase diminishes as more texts are included

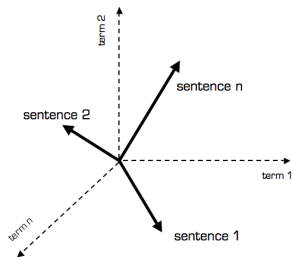


Geometric model

Any corpus (i.e., a collection of n documents) can be represented in the vector space model by a document-term matrix

A vector space model is a basic modeling mechanism for a word- or document-space (whether we look at rows or columns)

- ▶ A document vector with only one word is collinear to the vocabulary word axis
- ▶ A document vector that does not contain a specific word is orthogonal/perpendicular to the word axis
- ▶ Two documents are identical if they contain the same words in a different order



Document space	t_1	t_2	t_3	...	t_n	Term vector space
D_1	a_{11}	a_{12}	a_{13}	...	a_{1n}	
D_2	a_{21}	a_{22}	a_{23}	...	a_{2n}	
D_3	a_{31}	a_{32}	a_{33}	...	a_{3n}	
...						
D_m	a_{m1}	a_{m2}	a_{m3}	...	a_{mn}	
Q	b_1	b_2	b_3	...	b_n	